MxWithPandas

October 29, 2017

1 MX

```
In [1]: #%dhist
        #0: PT3S\nb (notebooks ipynbDir)
        #%cd -0
        #%cd ..
        # notebooks workingDir should be PT3S
```

```
In [2]: !pip uninstall PT3S -y
```

```
Uninstalling PT3S-0.0.16.dev1:
  Successfully uninstalled PT3S-0.0.16.dev1
```

```
In [3]: !pip install PT3S
```

```
Collecting PT3S
  Using cached PT3S-0.0.16.dev1-py3-none-any.whl
Installing collected packages: PT3S
Successfully installed PT3S-0.0.16.dev1
```

```
In [4]: from PT3S import Mx
```

```
In [5]: mx=Mx.Mx(h5File='./testdata/M-1-0-1.h5')
```

2 to Pandas

MX-Channel TIMESTAMP (1st MX-Channel) used as DataFrame Index; all other MX-Channels are DataFrame Columns

```
In [6]: import pandas as pd
```

the DataFrame

```
In [7]: df=mx.pdf
```

```
In [8]: type(df)
```

```
Out[8]: pandas.core.frame.DataFrame
```

```
In [9]: ts=df.iloc[1:-2,0]
        type(ts)
```

```
Out[9]: pandas.core.series.Series
```

Timestamps

```
In [10]: ts.index[0]
```

```
Out[10]: Timestamp('2017-06-18 23:00:00')
```

```
In [11]: ts.index[-1]
```

```
Out[11]: Timestamp('2017-06-19 00:30:00')
```

DatetimeIndex

```
In [12]: type(ts.index)
```

```
Out[12]: pandas.tseries.index.DatetimeIndex
```

Time Zone (tz_convert(None) will remove timezone after converting to UTC time)

```
In [13]: print(ts.index.tz)
```

```
None
```

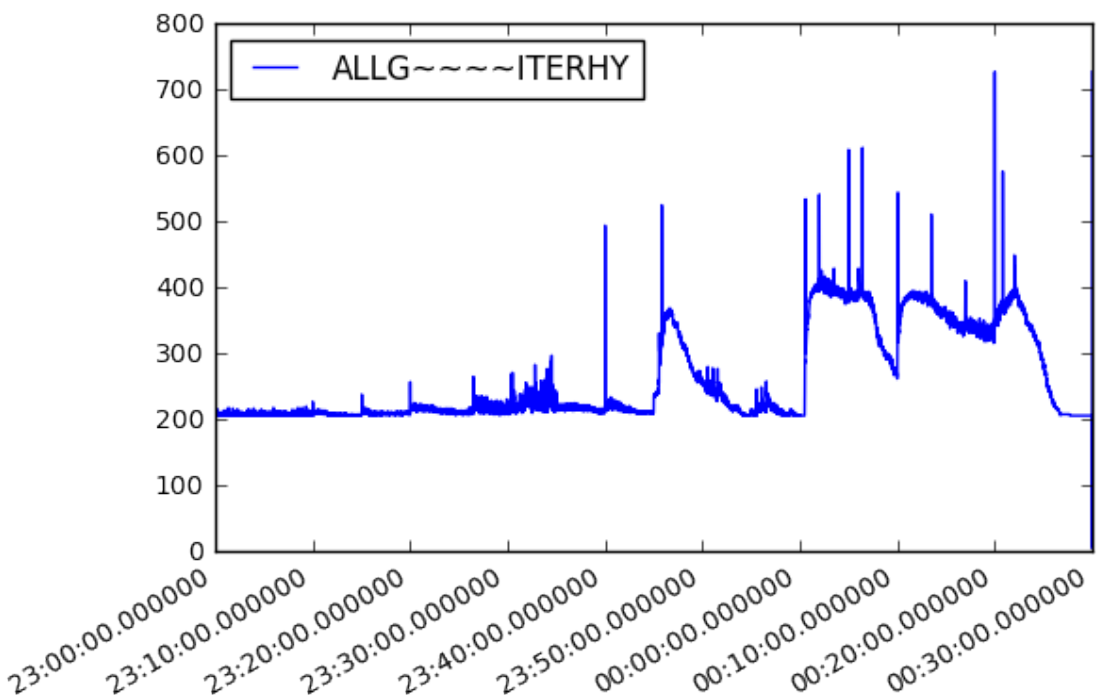
2.0.1 Time Series Plot

```
In [14]: %matplotlib inline
         import matplotlib.pyplot as plt
         from matplotlib import dates
```

by MX-Channel-Idx

```
In [15]: plt.figure()
         df.iloc[:,17].plot()
         plt.legend(loc='best')
```

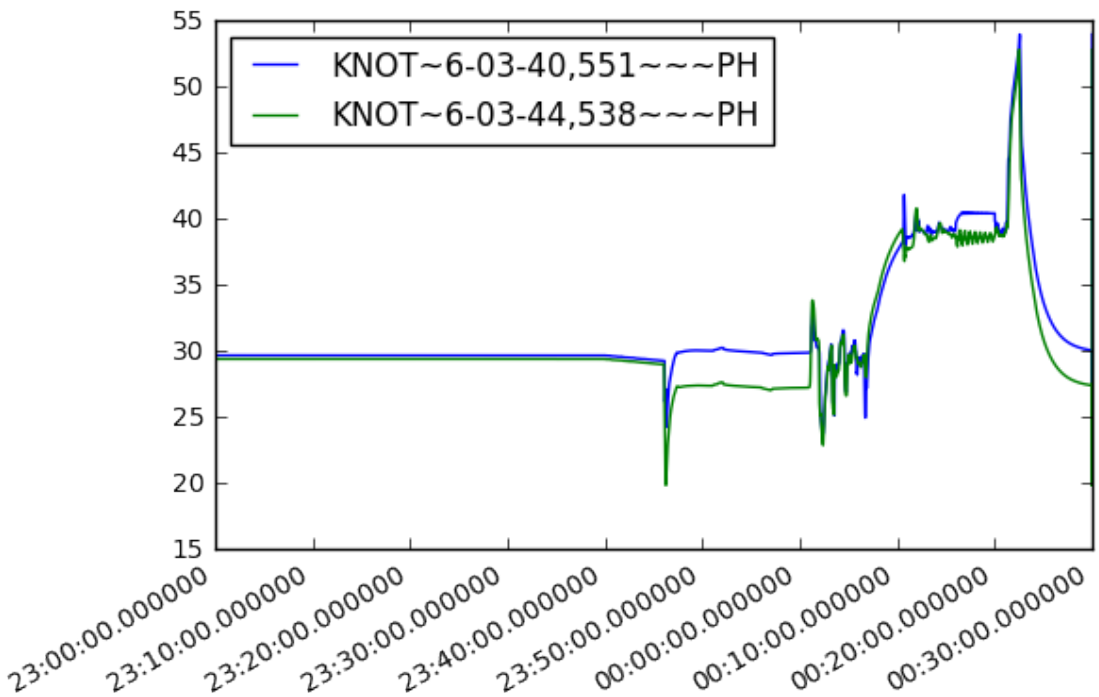
```
Out[15]: <matplotlib.legend.Legend at 0x252e49aeeb8>
```



by MX-Channel-Sir3sID

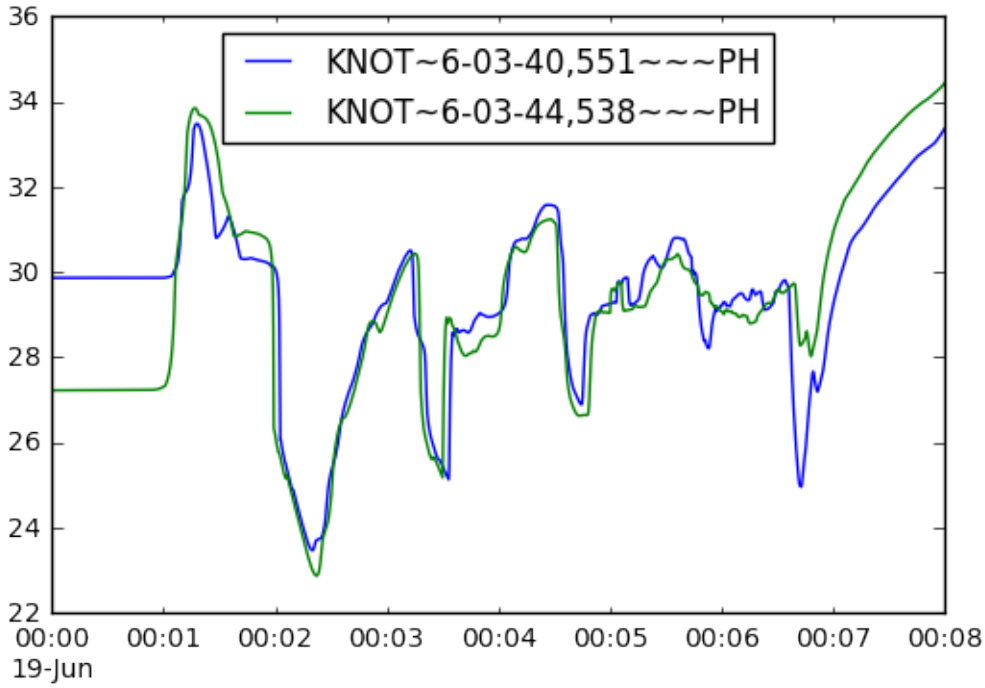
```
In [16]: import re
         regExpCompiled=re.compile(r'^KNOT~6-03-4\d+(\S+)~(\S*)~(\S*)~PH')
         df_colsel=df.loc[:,sorted([column for column in df.columns if
         regExpCompiled.search(column) != None])]
         df_colsel.plot()
```

```
Out[16]: <matplotlib.axes._subplots.AxesSubplot at 0x252e4d0a0b8>
```



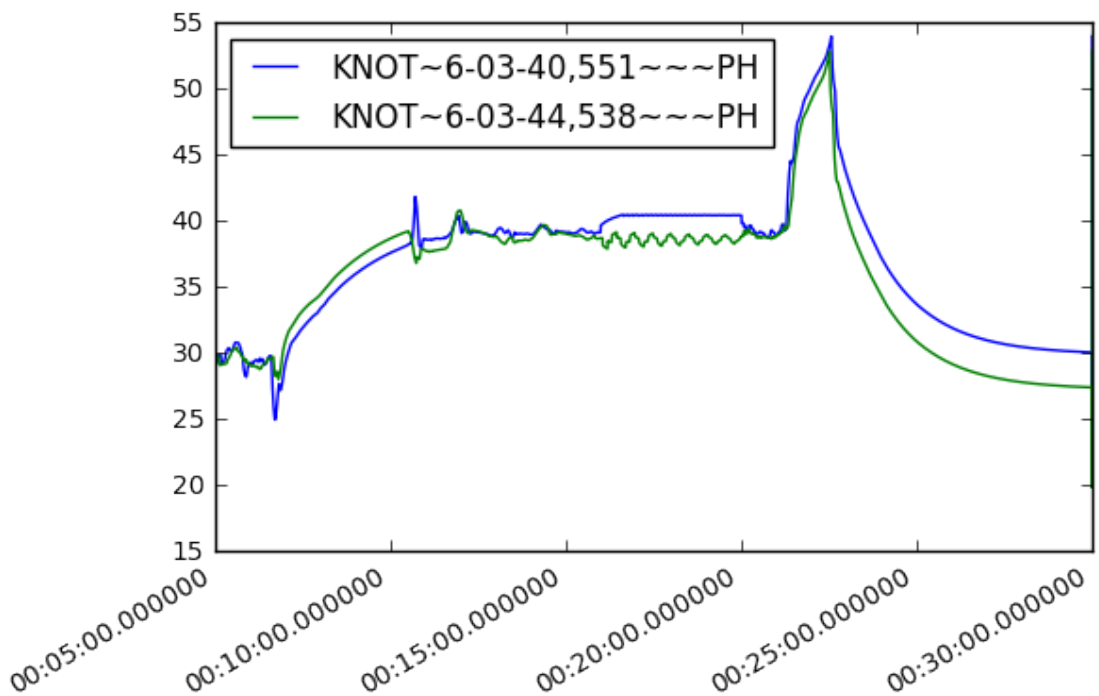

```
In [17]: start=df_colsel.index[0].to_datetime()+pd.DateOffset(minutes=60)
         end=start+pd.DateOffset(minutes=8)
         df_colsel.between_time(start_time=start.time(), end_time=end.time()).plot()
```

```
Out[17]: <matplotlib.axes._subplots.AxesSubplot at 0x252e516c828>
```




```
In [18]: df_colsel[pd.datetime(2017,6,19,0,5):].plot()
```

```
Out[18]: <matplotlib.axes._subplots.AxesSubplot at 0x252e50be668>
```



SnapShotType

```
In [19]: len(df)
```

```
Out[19]: 21604
```

```
In [20]: df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].head(3)
```

```
Out[20]:                          ALLG~~~~SNAPSHOTTYPE
         2017-06-18 23:00:00.000              b'STAT'
         2017-06-18 23:00:00.000              b'TIME'
         2017-06-18 23:00:00.250              b'TIME'
```

```
In [21]: df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].tail(2)
```

```
Out[21]:                      ALLG~~~~SNAPSHOTTYPE
         2017-06-19 00:30:00              b'TMIN'
         2017-06-19 00:30:00              b'TMAX'
```

```
In [22]: df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE',r'ALLG~~~~ITERHY']].groupby(r'ALLG~~~~SNAPSHOTTYPE').c
         ount()
```

```
Out[22]:                        ALLG~~~~ITERHY
         ALLG~~~~SNAPSHOTTYPE
         b'STAT'                             1
         b'TIME'                         21601
         b'TMAX'                             1
         b'TMIN'                             1
```

```
In [23]: df2=df[df[r'ALLG~~~~SNAPSHOTTYPE'].str.contains('TIME')]
```

```
In [24]: len(df2)
```

```
Out[24]: 21601
```

```
In [25]: df2.loc[:,'KNOT~6-03-44,538~~~PH'].head(4)
```

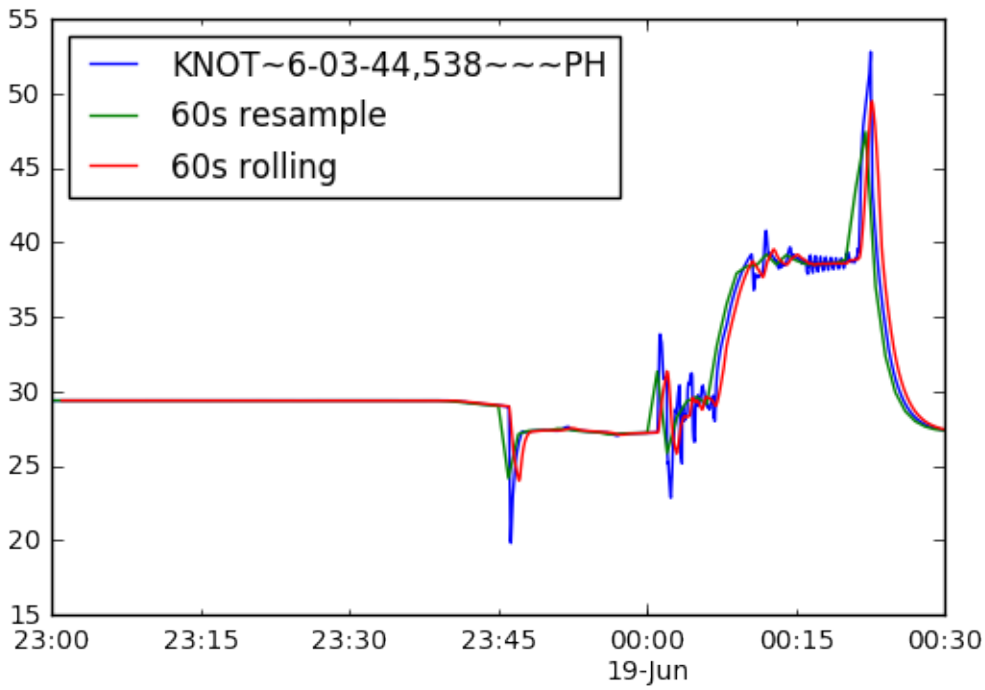
```
Out[25]: 2017-06-18 23:00:00.000    29.404100
         2017-06-18 23:00:00.250    29.404099
         2017-06-18 23:00:00.500    29.404100
         2017-06-18 23:00:00.750    29.404100
         Name: KNOT~6-03-44,538~~~PH, dtype: float64
```

2.0.2 Time Series Functions

```
In [26]: ts=df2.loc[:,'KNOT~6-03-44,538~~~PH']
```

```
In [27]: ts60=df2.loc[:,'KNOT~6-03-44,538~~~PH'].resample('60s').mean()
         ts60.name='60s resample'
         ts60r = ts.rolling(window=60*4).mean()
         ts60r.name='60s rolling'
         plt.figure()
         ts.plot()
         ts60.plot()
         ts60r.plot()
         plt.legend(loc='best')
```

Out[27]: <matplotlib.legend.Legend at 0x252e563a240>



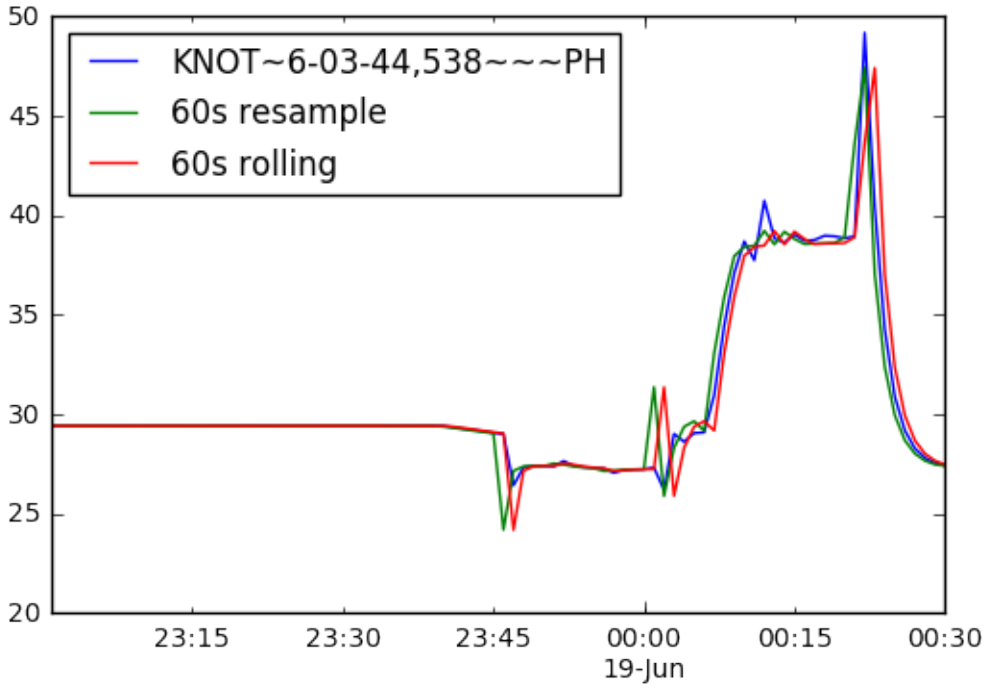

```
In [28]: df3=pd.DataFrame({ts60.name : ts60,ts.name : ts,ts60r.name : ts60r})
         df3=df3.reindex_axis([ts.name,ts60.name,ts60r.name], axis=1)
         df3.head()
```

Out[28]:

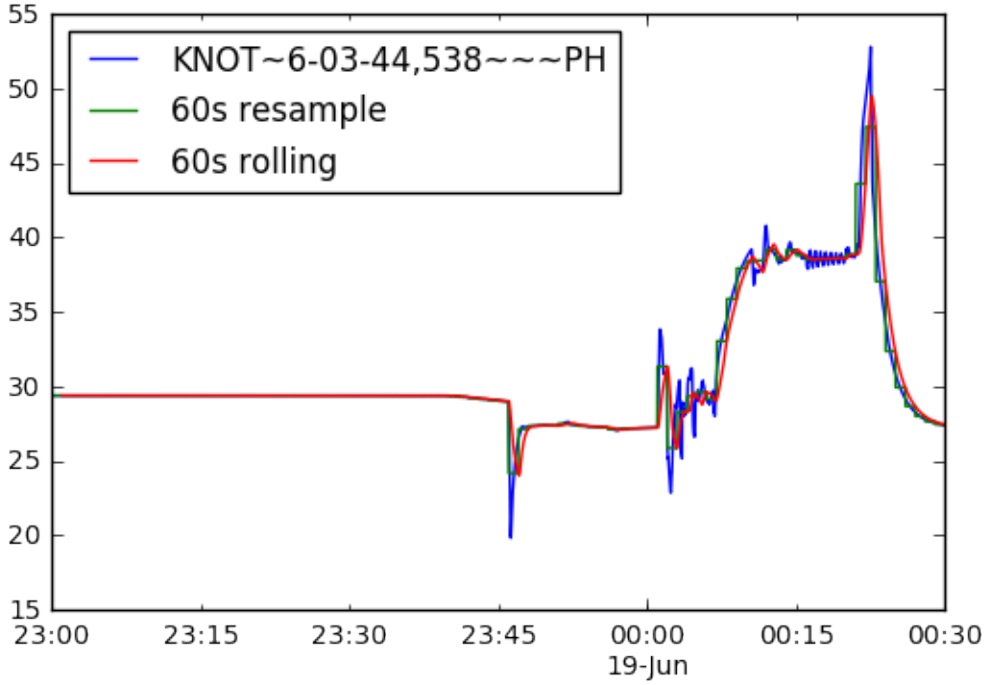
	KNOT~6-03-44,538~~~PH	60s resample	60s rolling
2017-06-18 23:00:00.000	29.404100	29.404088	NaN
2017-06-18 23:00:00.250	29.404099	NaN	NaN
2017-06-18 23:00:00.500	29.404100	NaN	NaN
2017-06-18 23:00:00.750	29.404100	NaN	NaN
2017-06-18 23:00:01.000	29.404099	NaN	NaN

```
In [29]: df3.dropna().plot()
```

Out[29]: <matplotlib.axes._subplots.AxesSubplot at 0x252e58e7208>




```
In [30]: pPlot = df3.ffill().plot()
```

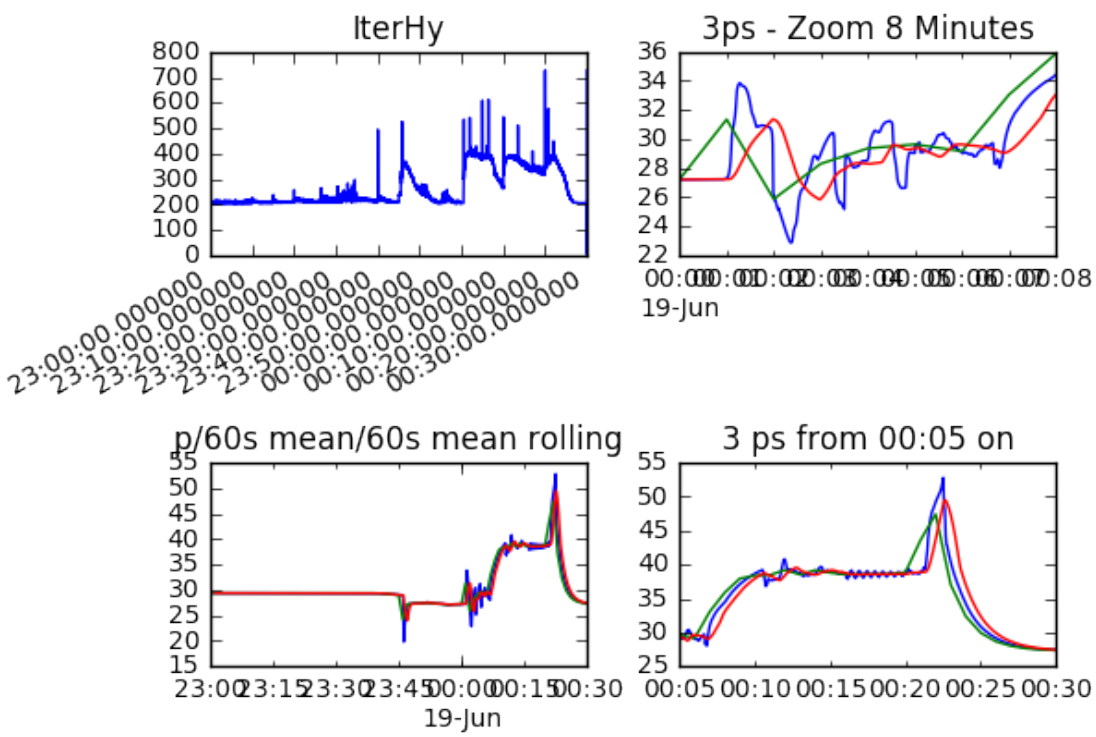


MATLAB, and pyplot, have the concept of the current figure and the current axes. All plotting commands apply to the current axes.

The function gca() returns the current axes (a matplotlib.axes.Axes instance), and gcf() returns the current figure (matplotlib.figure.Figure instance).

The subplot() command specifies numrows, numcols, num where num ranges from 1 to numrows*numcols.

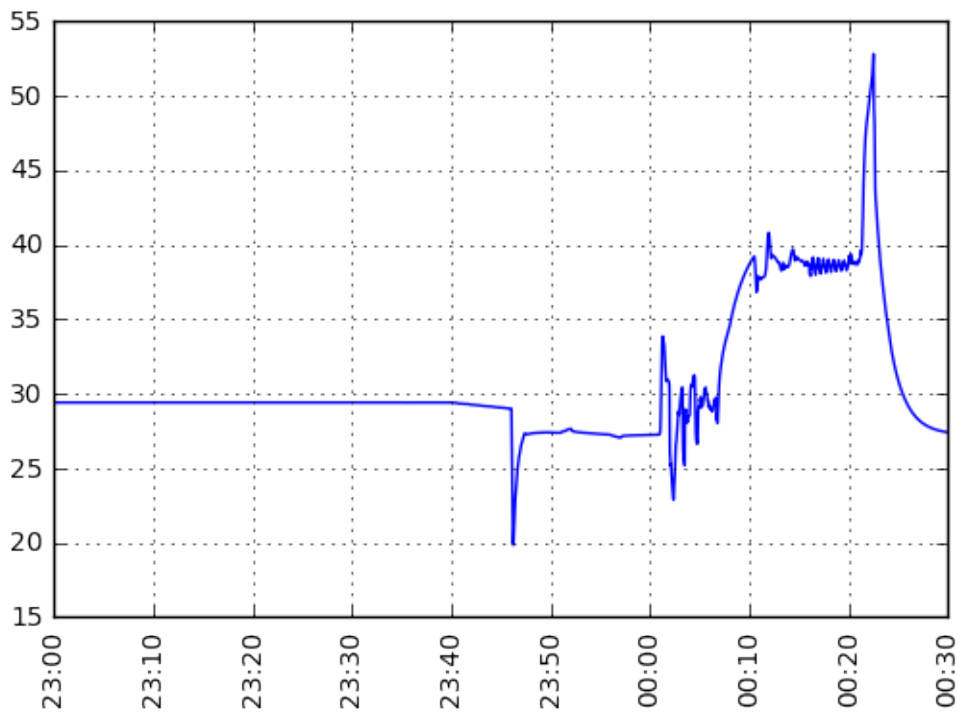
```
In [31]: plt.figure(1)

         plt.subplot(221)
         plt.title('IterHy')
         df.iloc[:,17].plot()

         plt.subplot(223)
         plt.title('p/60s mean/60s mean rolling')
         ts.plot()
         ts60.plot()
         ts60r.plot()

         plt.subplot(222)
         plt.title('3ps - Zoom 8 Minutes')
         ts.between_time(start_time=start.time(), end_time=end.time()).plot()
         ts60.between_time(start_time=start.time(), end_time=end.time()).plot()
         ts60r.between_time(start_time=start.time(), end_time=end.time()).plot()

         plt.subplot(224)
         plt.title('3 ps from 00:05 on')
         ts[pd.datetime(2017,6,19,0,5):].plot()
         ts60[pd.datetime(2017,6,19,0,5):].plot()
         ts60r[pd.datetime(2017,6,19,0,5):].plot()
         plt.tight_layout(pad=0.4, w_pad=0.5, h_pad=1.0) #plt.subplots_adjust(top=0.92, ...)

         plt.show()
```



```
In [32]: #fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(nrows=2, ncols=2)
         #example_plot(ax1)
         #def example_plot(ax, fontsize=12):
         #    ax.plot(...)
```

"Hand-made" xaxis

```
In [33]: plt.close('all')
         plt.figure()
         plt.subplot()
         plt.gca().xaxis.set_major_locator(dates.MinuteLocator(interval=10))
         plt.gca().xaxis.set_major_formatter(dates.DateFormatter('%H:%M'))
         #plt.gcf().autofmt_xdate()
         plt.plot(ts.index, ts)
         labels=plt.gca().get_xticklabels()
         plt.setp(labels,rotation=90)
         plt.grid()
         plt.show()
```



3 PIP

```
In [34]: !pip install -e .
```

```
Obtaining file:///C:/Users/Wolters/Source/Repos/PT3S
Installing collected packages: PT3S
  Found existing installation: PT3S 0.0.16.dev1
```

```
    Uninstalling PT3S-0.0.16.dev1:
      Successfully uninstalled PT3S-0.0.16.dev1
  Running setup.py develop for PT3S
Successfully installed PT3S
```